

Baseline data collection
2006
2010
Imaging Data Collection
2014
2024
Prediction Window
First Imaging Visit
2024

ukbiobank

Medical Imaging

Genetics

ECG and Clinical

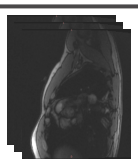
3D Cardiac MRI Data

2D MRI and DXA Data

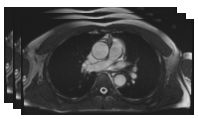
Genotype Data



Long Axis



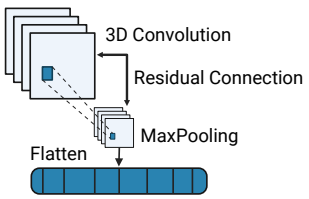
Blood Flow

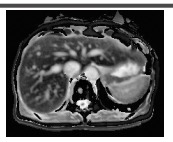


Aortic Distensibility

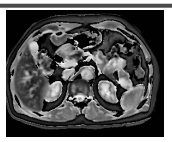
ResNet-18 MedicalNet pretraining

80,000 3D volumes

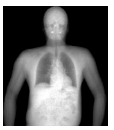




Liver MRI



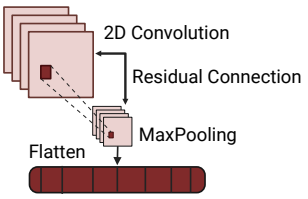
Pancreatic MRI



Whole body DXA

ResNet-50 RadImageNet pretraining

1.35M 2D radiology images



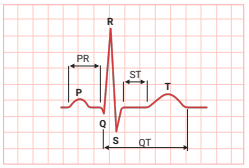
FINNGEN

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GWAS summary stats from Biobanks

METAL

229k cases and 1.03M controls



Machine derived ECG features



LDL, HDL, Triglycerides, Total Cholesterol, apoA, apoB

Blood Biochemistry



Age, Sex, BMI, Smoking Status, Diabetes Status, Socioeconomics, Medication History

Embedding(s)

Embedding(s)

PRS-CS

PCA

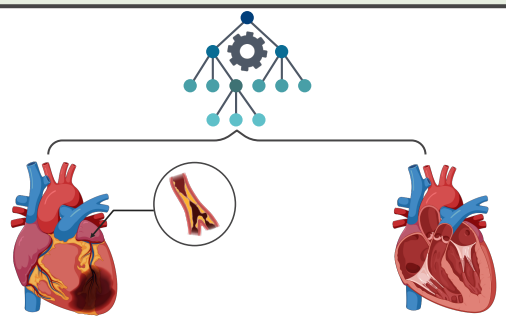
PCA

PRS CAD

Demographic and lifestyle data

Tabular Features (spatially cross validated train and test datasets)

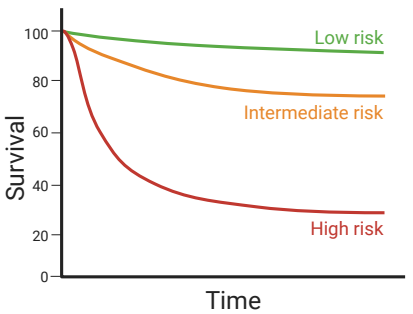
Tree Based Models for CAD classification



Diseased

Healthy

Cox Proportional Hazards Model for CAD incidence



Risk Stratification